

Improving English Listening Skills in Japanese Learners with Speech Rhythm Vibration Feedback

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Abstract

We investigated the use of rhythmic vibration feedback to improve English listening skills in Japanese learners, specifically targeting the challenge of perceiving and processing the stress-timed rhythm of English. Three experiments explored the impact of synchronized vibration patterns, corresponding to stressed and unstressed syllables, on prosody perception, recognition of compound words and noun phrases, and word anticipation. Results indicate that the combined voice and vibration feedback significantly improved the recognition of stressed and unstressed syllables and enhanced word prediction compared to voice alone. However, no effect on compound word recognition was observed. This research suggests that rhythmic vibration feedback can be a valuable tool for enhancing English listening skills in Japanese learners by facilitating the acquisition of stress-timed rhythmic patterns.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; Ubiquitous and mobile computing systems and tools; Human computer interaction (HCI); Empirical studies in HCI.

Keywords

Speech entrainment, vibration feedback, language prosody, wearables

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1 Introduction

English as a Foreign Language (EFL) is taught worldwide, and unsurprisingly, there is a diverse range of tools available to support English learners. These tools range from large online platforms [43], artificial intelligence tools that aid in writing [38], enable conversation practice [13, 40], or provide feedback for oral presentations [30] as well as robot tutors teaching the language [11, 35]. In the domain of foreign language acquisition, listening is a crucial

skill that is essential yet challenging to master. While many tools support reading comprehension [5, 8, 12, 34], those designed for listening comprehension primarily focus on speech recognition by converting speech to text [3, 10, 18]. While some approaches acknowledge the importance of prosodic structure (understanding stress, intonation, and rhythm), they tend to focus on correcting pronunciation rather than directly enhancing speech perception [2, 28]. Although these tools are useful, they do not inherently improve listening skills. English, as a stress-timed language, depends on the correct pronunciation of both stressed and unstressed (or weakened) syllables to create its characteristic rhythm. For non-native speakers, mastering this rhythmic pattern is essential. This challenge is especially significant for Japanese learners, who often struggle to develop a native-like lexical stress pattern [16]. Based on the theory of neuronal entrainment, which highlights the importance of listeners synchronizing with the speaker's speech rhythm [6, 19], we developed a syllable-timed rhythmic vibration feedback method. This feedback was aligned with the stress-timed speech rhythm of English to enhance the speech rhythm perception of EFL learners. The inherent time-domain characteristics of vibrations, including frequency, amplitude, and duration, make them particularly suited for transmitting wave signals such as natural language [20]. In three exploratory experiments, we investigated the effects of rhythmic vibration feedback on syllable stress patterns, the speech timing of compound nouns and noun phrases, and word prediction. The findings suggest that providing rhythmic vibration feedback on the stressed and unstressed syllables of English words and phrases aids Japanese learners in accurately recognizing stress patterns and enhances word prediction. The contribution of our work are threefold: 1) We explore a new modality for language learning by demonstrating the potential of haptic-vibrotactile feedback as a supplementary channel for conveying prosodic information, which could be particularly beneficial for learners struggling with auditory processing; 2) Our findings highlight the importance of rhythmic processing in language comprehension suggesting that haptic feedback can facilitate the acquisition of stress-timed patterns; and 3) Our study's methodology and experimental setup can serve as a starting point for developing more sophisticated and integrated haptic interfaces for language education.

2 Related Work

2.1 Challenges of the Stress-timed Rhythm of English as Foreign Language

English is a stress-timed language, meaning its rhythm relies on the correct pronunciation of both stressed and unstressed (or weakened) syllables. For non-native speakers, understanding and mastering

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this rhythm is essential. Even if a speaker has accurate pronunciation, placing stress on the wrong syllable can hinder comprehension and significantly affects the intelligibility of speech. Common challenges for language learners involve prosodic features such as 1) duration (speech rate and timing), 2) intensity (amplitude and loudness), and 3) fundamental frequency (pitch) [42]. Content words such as nouns, verbs, adjectives, adverbs, and pronouns have the most stress in English and are important to express the main meaning of the sentence. To effectively learn English, foreign language learners should be introduced to the language's rhythm as early as possible, prioritizing recognition before production. They must grasp the stress-timed nature of English and how stress, connected speech, and intonation work together [31, 42]

Japanese learners often struggle with prosodic features such as stress, intonation, and rhythm of English and even Japanese learners with high English proficiency have difficulties to develop a native like stress pattern [16]. The mora-timed rhythm of the Japanese language, which lacks distinct stressed and unstressed syllables, is believed to pose a significant obstacle [16, 22] when acquiring the rhythm of a stress-timed language like English. A mora is a unit of sound that measures the length of words of utterances. Previous studies have shown that Japanese use mora for auditory perception and speech segmentation [32].

2.2 Neuronal Entrainment of Speech Rhythm and Attention

Speech exhibits temporal patterns that create a perceivable rhythm. In English, this rhythm is characterized by a sequence of stressed and unstressed syllables. Research has indicated that speech generates a rhythmic brain response that aligns with the pattern of the speech input, as demonstrated by [6]. This neuronal entrainment, where neural oscillations in the brain synchronize with the rhythm of the speech signal, is essential for speech perception [45] and plays a critical role in speech intelligibility [9, 29]. Entrainment of the speech input is triggered by elements of its linguistic structure, such as slow amplitude modulations in the temporal envelope of the speech signal [1, 15, 26]. Studies on children with reading difficulties indicate a connection between auditory processing and sensitivity to prosody [7]. Moreover, neural entrainment enhances the predictability of speech, aiding us in processing what we hear more effectively [17, 45]. Research by [4] demonstrated that concurrent stress patterns of word primes activate lexical access, highlighting the predictive value of recognizing prosodic structure. In summary, prior research suggests that the ability to process the rhythm of speech enables our brain to synchronize with these rhythms, thus enhancing our comprehension and our capacity to anticipate words.

3 Preliminary Testing of Haptic-Vibrotactile Feedback

Haptic-vibration feedback has demonstrated its effectiveness in various applications such as skill learning [25], as feedback for sequential gestural inputs [39], or for navigation tasks [44]. Moreover, its inherent properties such as frequency, amplitude, and duration make it a suitable alternative or enhancement for sound. For instance, haptic-vibration feedback derived from sound has been

utilized in sound art exhibitions [21], virtual reality applications for people who are deaf or hard-of-hearing [27, 36], and to enhance gaming experiences [41]. However, transforming speech signals into recognizable haptic feedback remains challenging and generally necessitates training to interpret the haptic signals [33]. As an initial step, we concentrated on creating perceivable rhythmic vibration feedback to help EFL learners recognize prosody, using a simple design that minimizes hardware demands, is portable, and requires minimal training. We used a compact coin vibration motor with a diameter of 8 mm, easily integrable into wearables. Vibrations can be generated from any audio source, like a laptop or smartphone, with adjustable intensity.

In a series of pretests with ten participants, we determined the vibration frequency, duration of vibration bursts, and vibration amplitude. The goal of the pretests was to establish parameter settings that would create an easily perceivable vibration rhythm, consisting of strong and weak vibration bursts. We found that vibrations at 150Hz were the most easily perceivable. We also discovered that very short vibration bursts were quite uncomfortable. However, to perceive a vibration rhythm—a distinguishable pattern of vibration bursts over time—each burst needed to be followed by a silent interval, preferably 100 msec. Considering a natural speech rate of around 5Hz [24], we decided on a basic pattern of a 100 msec vibration burst followed by a 100 msec silence interval, which was then adjusted and synchronized with the syllable rhythm. Additionally, it was crucial for strong and weak vibration bursts to be clearly distinguishable. The pretests revealed that weak bursts should have an amplitude 75% lower than that of strong vibration bursts. Figure 1 displays how vibration feedback was generated for stressed and unstressed syllables in Experiment 1 (left), the temporal rhythm of compound nouns and noun phrases in Experiment 2 (middle), and for word anticipation in Experiment 3 (right).

4 Overall Study goals and Hypotheses

The goal of the study was to support Japanese EFL learners to recognize the stress-timed speech rhythm of English by providing rhythmic vibration feedback for intensity changes (intonation) of words and phrases as well as for speech rate and timing. In three small experiments we explored if syllable timed rhythmic vibration feedback of stressed and unstressed syllables improves perception of prosody and word prediction. The following three hypotheses were tested:

- H1: It is anticipated that rhythmic vibration feedback of strong and weak vibrations for stressed and unstressed syllables enhances perception of speech prosody.
- H2: It is expected that rhythmic vibration feedback of the temporal distribution of syllables improves recognition for compounds and noun phrases.
- H3: It is expected that rhythmic vibration feedback of strong and weak vibrations for stressed and unstressed syllables improves word anticipation when listening to word fragments.

Voice stimuli for all experiments were created with Google Cloud Text-to-Speech API using the female voice en-US-Wavenet=F. WaveNet models generate a more natural speech with human-like emphasis, inflection on syllables, phonemes and words. Consent

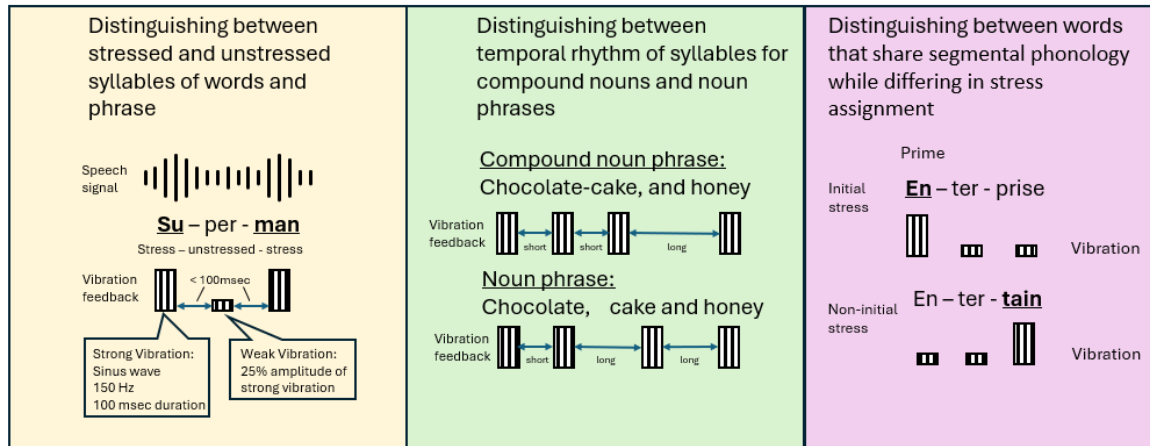


Figure 1: Vibration generation methods. Left: vibration feedback for stressed and unstressed syllables in Experiment 1. Middle: Vibration feedback for temporal distribution of syllables for compound nouns and noun phrases in Experiment 2. Right: Vibration feedback for stressed and unstressed syllables for word anticipation in Experiment 3.

was obtained by all participants at the beginning of each experiment. The ethical guidelines of Kyocera as well as the ethical guidelines according to the declaration of Helsinki for research involving Human Subjects were maintained. Participation was voluntary and informed consent was obtained from all participants. All participants were of normal hearing. Collected data were tested for normal distribution and when assumptions were violated a nonparametric test was used.

5 Experiment 1: Vibration feedback for intonation patterns

Experiment 1 explored whether perception of the prosodic structure could be improved with synchronized rhythmic vibration feedback of strong and weak vibrations for stressed and unstressed syllables (H1).

5.1 Method

5.1.1 Experimental Design and Procedure. Eight participants (two females, ages 25–60) participated in a within-subject experiment experiencing three conditions: 1) voice only, 2) vibration only, and 3) voice+vibration. In the voice only condition, English words and phrases were delivered via headphones. For the vibration only condition, vibrations representing the English words and phrases were provided through a small wrist worn vibration device. In the voice+vibration condition, participants received English words and phrases with synchronized stress-timed vibration feedback. Two sets of stimuli were used: one presented across all three conditions and the other only in the voice+vibration condition. This setup was designed to assess learning effects and compare performance between the voice only and voice+vibration conditions. The order of stimuli sets was counterbalanced, and items within each set were presented randomly.

5.1.2 Experimental Stimuli.

- **Voice stimuli:** Experimental stimuli were taken from the DEEdie task used by [7, 23]. Two stimuli sets, each containing 21 words and phrases were selected. Sets were balanced for total number of syllables (72 each) as well as word and phrase length ranging from two to five syllables per item.
- **Vibration Stimuli:** Parameter settings for the vibrations were based on pretest results as reported in section 3. Vibration bursts were synchronized to the timing of syllables using the spectral pattern of the voice stimuli. Stressed and unstressed syllables were identified based on the DEEdie pattern indicated by [7, 23]. See also Figure 1 for reference.

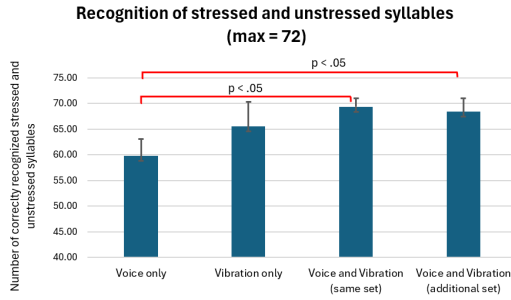
5.2 Results and Discussion

The Wilcoxon test for paired samples revealed that participants identified more stressed and unstressed syllables correctly in the voice+vibration condition ($Md = 70.5$) compared to the voice only condition ($Md = 59.5$) for the same item set ($V = 2$, $z = -1.88$, $p = 0.03$) as well as for the additional item set ($Md = 70$), $V = 1$, $z = -1.82$, $p = 0.03$. No differences were found between voice only ($Md = 70.5$) and vibration only ($Md = 71$) condition ($V = 8$, $z = -0.90$, $p = .18$) and between the two items sets (same set and additional set) in the voice+vibration condition ($V = 13.5$, $z = 0.25$, $p = 0.59$), see also Figure 2.

The results indicated that participants had a higher success rate in distinguishing between stressed and unstressed syllables when vibration feedback was provided with voice stimuli. This was true for both practiced and new sets of items. Vibration patterns alone were detected at a rate similar to voice alone, showing that participants could differentiate between strong and weak vibrations. These findings suggest that vibration feedback on syllable stress improved the perception of English stress-timed patterns for Japanese learners, potentially assisting non-native speakers in adapting to the rhythm of spoken English, a necessary prerequisite to learn the language effectively [42].

Table 1: Average recognition rate of compound nouns and noun phrases with and without vibration and comparison of recognition rate for the first 20 items and the last 20 items

	with Vibration (max 20)	without Vibration max 20)	1st 20 items	last 20 items
Mean	15.67	15.00	13.67	17.00
Median	16.00	15.00	13.00	17.00
SD	2.50	1.73	2.87	1.12

**Figure 2: Recognition rate of stressed and unstressed syllables (maximum score = 72)**

6 Experiment 2: Vibration feedback of speech timing

Experiment 2 explored whether vibration feedback on speech timing improves recognition of compounds (e.g. chocolate-cake) and noun phrases (e.g. chocolate, cake) in a dichotomic listening task (H2). Vibration feedback was synchronized with the syllable pauses of the target phrases.

6.1 Method

6.1.1 Experimental Design and Participants. Nine participants (one female, aged 26–65) participated in the study. Using a within-subject design, the study examined the effect of vibration feedback on syllable timing. Participants engaged in a dichotic listening task where pairs of compound nouns and noun phrases (e.g., chocolate-cake and honey vs. chocolate, cake, and honey) were simultaneously played through headphones. Voice stimuli were randomly paired with or without vibration feedback, whereby vibration feedback was always synchronized to the syllable timing of the voice played in the left ear. Participants were instructed to report only the voice stimuli played on the left ear and recognition accuracy was measured.

6.1.2 Experimental stimuli.

- Voice stimuli: Twenty pairs of compound nouns and noun phrases from the study by [37] were used in the experiment.
- Vibration stimuli: Vibration bursts with a frequency of 150Hz and duration of 100 msec were created. Vibration bursts were synchronized to the timing of syllables and pauses. See also Figure 1 for reference.

6.2 Results and Discussion

A Wilcoxon test revealed no significant differences in recognition rates for items with and without vibration feedback ($V = 26$, $z = -.56$, $p = .28$). A comparison of recognition rates between items at the beginning of the trial (first 20 items) and items at the end of the trial (last 20 items) revealed a significant increase in recognition rate for the last items indicating a learning effect; $V = 1$, $z = -2.05$, $p = .02$. Mean, Median and Standard Deviation are displayed in Table 1.

Vibration feedback did not result in an improved recognition rate for compound nouns and noun phrases. Participants' feedback indicated that due to the dichotomic listening task, they focused their attention entirely on the voice played at the left ear and rather ignored the vibration feedback when provided. Furthermore, recognition rate increased throughout the trial indicating that participants became better in focusing their attention on the left side while ignoring the voice stimuli played at their right side. It is also believed that participants became better in identifying compound nouns and noun phrases based on the syllable timing throughout the trial resulting in a nearly perfect score for the last 20 items of the trial.

7 Experiment 3: Vibration feedback for Word anticipation

The third study examined whether rhythmic vibration feedback of strong and weak vibrations for stressed and unstressed syllables improves word anticipation when listening to word fragments (H3). Words that share segmental phonology while differing in stress assignment (Admiral – Admiration) were used in a priming task.

7.1 Method

7.1.1 Experimental Design and Procedure. A total of thirteen participants (2 females, age range 27–60) took part in the study. A within subject design with the factor with/without vibration feedback was administered in the study. Pairs of target primes that shared segmental phonology while differing in stress assignment (e.g. admir... → admiral / admiration) were visually presented together with a distractor. Participants listened to the prime and had to select the word associated with the prime. Two sets of spoken primes (each set containing 24 stimuli) were presented in two blocks with or without vibration feedback of the stress-timed rhythm of the target word. The two sets of spoken primes and the two trial blocks were counterbalanced. Recognition rate for the target prime was measured. At the end of each set participants answered questions about task difficulty on a 7-point Likert scale ranging from 1-not difficult at all till 7-very difficult.

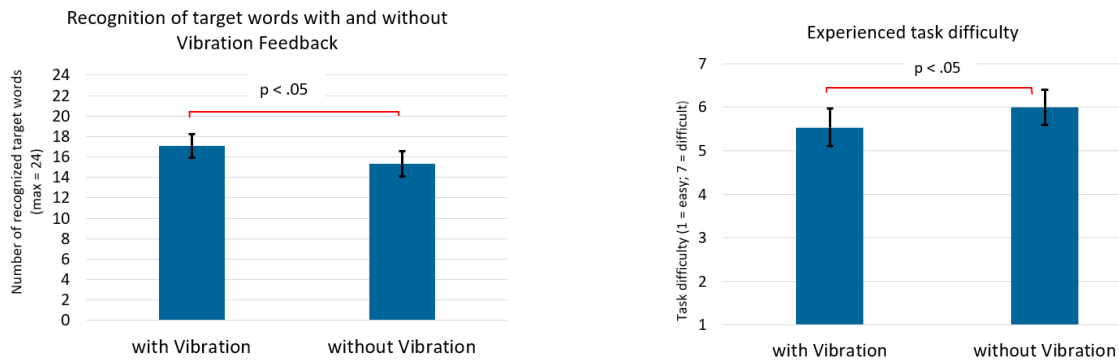


Figure 3: Left: Recognized target words in condition with and without vibration feedback (max = 24). Right: Experienced task difficulty in condition with and without vibration feedback.

7.1.2 Experimental stimuli.

- **Voice stimuli.** Twenty-four word pairs (48 words) that shared segmental phonology while differing in stress assignment were selected from [23]. Voice stimuli were edited with the audio editor Audacity to retain only the first two syllables of each word. Three experimenters then tested these primes to ensure they were neither too easy nor too difficult to recognize the target word. The twenty-four word pairs were divided into two balanced prime sets, each containing 24 word primes.
- **Vibration stimuli.** The vibrations for stressed and unstressed syllables were generated in the same manner as in Experiment 1. Stressed and unstressed syllables were identified based on the stress pattern indicated by [23]. See also Figure 1 for reference.

7.1.3 Results and Discussion. A paired sampled t-test revealed that participants recognized more target words correctly with ($M = 17.08$, $SD = 2.36$) than without vibration feedback, $M = 15.21$, $SD = 2.46$; $t(12) = 2.27$, $p = .04$. Furthermore, task difficulty was evaluated as more difficult in the condition without vibration ($M = 6.00$, $SD = 0.82$) than with vibration, $M = 5.54$, $SD = 0.88$; $V = 0$, $z = -2.06$, $p = .02$ (see Figure 3).

The results indicate that vibration feedback for target words of voice primes increased recognition rate. The results suggest that supporting listeners in recognizing the syllable-timed stress pattern of target words when only being able to hear partials of the words might improve prosodic processing and improves word prediction. These can be especially useful in situations where speech perception is impaired due to noise or in multitalker situations. Differences in experienced task difficulty indicate that vibration feedback supported the listening task. The rhythmic vibration feedback might increase alertness as well as guiding attention to relevant parts of the word (stressed syllables) that aided in word identification.

8 Conclusion

The study reported here investigated the use of rhythmic vibration feedback to improve English listening skills in Japanese learners, focusing on the challenges posed by the stress-timed rhythm of

English [14, 15]. The novelty lies in applying vibrotactile feedback, not for pronunciation correction (as is common), but to enhance the perception of English prosody, leveraging the principle of neuronal entrainment [6, 19]. This approach directly addresses the difficulty Japanese learners face in distinguishing stressed and unstressed syllables, a crucial aspect of English comprehension [42]. The paper explored the impact of this feedback method on three key areas: perception of intonation patterns, recognition of compound words/phrases, and word prediction.

The results of three experiments demonstrate the potential of haptic-vibrotactile cues to improve perception of intonation patterns and enhance word anticipation. While further investigation is needed to fully understand the impact on compound word/phrase recognition, the initial findings are promising. This work presents a novel application of vibrotactile feedback in language learning, opening possibilities for designing more effective and engaging learning tools, particularly for learners struggling with prosodic aspects of a foreign language. Future research should explore the integration of this method into different learning environments, and its potential for learners of other languages with rhythmic speech patterns distinct from their native tongue. Additionally, exploring the optimal combination of vibrotactile feedback with auditory and visual cues could further enhance its effectiveness in improving listening comprehension and overall language acquisition.

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APPENDICES

A sets 1 and 2 for the DEEdEE task of Experiment 1

List of words and word phrases used in Experiment 1. The syllables in bold print indicate stressed syllables.

Set 1	Set 2
Tony Blair	Sleeping Beauty
Brad Pitt	Mary Poppins
Bill Clinton	Independence Day
Aladdin	The Simpsons
Home Alone	The Muppets
Dracula	Snow White
Goodfellas	Toy Story
The Snow Queen	Chicago
Godzilla	Harry Potter
Total Recall	Superman
Humpty Dumpty	Beyonce
The Lion King	Sesame Street
Bambi	David Beckham
Pokémon	Lost in Space
Jack and Jill	Peter Pan
Winnie the Pooh	Chinatown
Pinocchio	The Saddle Club
Bob the Builder	The Jungle Book
Thumbelina	Peter Kay
Planet of the Apes	Back to the Future
Wallace and Gromit	Kylie

B Stimuli for Compound Nouns and Noun Phrases task of Experiment 2

List of compound nouns and corresponding noun phrases used in Experiment 2

Compound nouns	Noun phrases
chocolate-cake and honey	chocolate, cake and honey
twenty-one and six	twenty, one and six
football and socks	foot, ball and socks
paperbag and string	paper, bag and string
bowtie and shoes	bow, tie and shoes
fruit-salad and milk	fruit, salad and milk
beanbag and flowers	bean, bag and flowers
sunlight and tress	sun, light and trees
breadstick and eggs	bread, stick and eggs
paintbrush and water	paint, brush and water

C Stimuli for Word Prediction Task of Experiment 3

Primes with initial stress and non-initial stress and distractors used in Experiment 3.

Prime (Initial stress)	Prime (Non-initial stress)	Distractor
Admiral	Admiration	Mosquito
Analogue	Analytic	Compensation
Animal	Anniversary	Proportion
Arrogant	Aromatic	Generous
Ceremony	Cerebellum	Permission
Compromise	Comprehend	Discipline
Conference	Confirmation	Manipulate
Consequence	Conservation	Obnoxious
Corridor	Correspond	Invention
Diagram	Diabetes	Apology
Enterprise	Entertain	Foundations
Etiquette	Etymology	Volcano
Exercise	Exhibition	Messenger
Horrible	Horizontal	Reputation
Immigrant	Immature	Catastrophe
Impotent	Impolite	Reflection
Interval	Interfere	Residence
Manicure	Manifestation	Accelerate
Metaphor	Metamorphosis	Seriously
Motorbike	Motivation	Umbrella
Opera	Opposition	Encouragement
Prominent	Promenade	Illusion
Property	Propaganda	Hesitation
Repertoire	Repetition	Initiative